PAKISTAN TECH, ENERGY & MARITIME MASTER REGISTRY

Advanced Dossier Module V4: Naval Fleet Architecture, Civil Nuclear Energy Infrastructure, Space Programs, and Digital Tech Ecosystems.

1. STRATEGIC NAVAL & MARITIME FLEET ARCHITECTURE

Pakistan's maritime strategy revolves around protecting the Sea Lines of Communication (SLOC) in the North Arabian Sea, securing Gwadar deep-sea approaches, and maintaining a credible sea-based second-strike capability.

Asset Class / Program	Origin / Partners	Core Technical Specifications & Deployment Role
Hangor-Class Submarines (Project 039A/B Base)	China / Domestic (KS&EW;)	8 diesel-electric attack submarines equipped with Air-Independent Propulsion (AIP) for extended underwater endurance. First 4 built in China, remaining 4 being built at Karachi Shipyard & Engineering Works (KS&EW;). Engineered to deploy Babur-3 sub-surface cruise vectors.
Tughril-Class Frigates (Type 054A/P)	China (Hudong-Zhonghua)	4 guided-missile stealth frigates forming the surface backbone of the PN fleet. Loaded with advanced anti-air (LY-80N) and supersonic anti-ship / land-attack cruise missiles (CM-302).
Milgem Class / Babur-Class Corvettes	Turkey (ASFAT) / Domestic	4 multi-mission stealth corvettes. Features local combat management suites (Harbah-NG weapon systems) and advanced anti-submarine warfare (ASW) capabilities.
Jinnah-Class Frigates	Turkey / Pakistan Joint Design	Indigenous indigenous frigate development program program following technology transfer from the Milgem initiative, optimizing local steel manufacturing and defense integration.

2. CIVIL NUCLEAR POWER & ENERGY LANDSCAPE

Regulated by the Pakistan Nuclear Regulatory Authority (PNRA) and operated by the Pakistan Atomic Energy Commission (PAEC), the civil nuclear infrastructure plays a central role in mitigating base-load energy supply deficits.

Facility / Station	Active Reactors	Net Capacity	Reactor Technology & Strategic Operational Context
Karachi Nuclear Power Plant (KANUPP)	K-2, K-3	2,200 MW total	Utilizes advanced HPR1000 (Hualong One) pressurized water reactors generation III+ designs. Features robust passive safety configurations and double containment layers.
Chashma Nuclear Power Plant (CHASNUPP)	C-1, C-2, C-3, C-4	1,330 MW total	4 operational CNP-300 pressurized water reactors. Chashma-5 (C-5) is currently in pipeline planning stages to add a 1,200 MW Hualong One unit.

3. CIVILIAN SPACE PROGRAM & SATELLITE CONSTELLATIONS

SUPARCO (Space & Upper Atmosphere Research Commission) handles state space research, remote sensing, and orbital asset deployment.

Satellite / Asset	Launch Date	Orbit Type	Primary Technical & Mission Purpose
PakSat-1R	2011	GEO (38° E)	Communications satellite delivering cross-continental broadband, TV broadcasting, emergency telecommunications, and digital routing arrays.
PRSS-1 (Pakistan Remote Sensing Satellite)	2018	SSO (Sun-Synchronous)	High-resolution optical earth observation platform utilized for geospatial mapping, environmental monitoring, and border surveillance tasks.
PakTES-1A	2018	LEO (Low Earth)	Indigenously co-developed technology evaluation satellite verifying local sensor systems, tele-command suites, and satellite bus designs.
iCube-Qamar	May 2024	Lunar Orbit	Pakistan's historic deep-space cubesat orbiter launched via Chang'e 6 mission, imaging lunar surface grid matrices and acquiring deep-space particle telemetry.

4. DIGITAL ECOSYSTEM, SOFTWARE ENGINEERING & FINTECH

Bypassing old agricultural indicators, Pakistan's tech landscape is scaling swiftly across IT export remittance, localized software architectures, and automated digital finance engines.

Sector Domain	Key Structures / Drivers	Ecosystem Significance & Growth Parameters
IT Exports & Freelance Networks	PSEB / Freelance Infrastructure	Ranked as a top global provider for software development, cloud engineering, and technical architecture on major platforms, supported by Pakistan Software Export Board (PSEB) special tech zones.
FinTech Infrastructure	Raast Payment Gateway / SBP	State Bank of Pakistan's (SBP) instant, zero-cost digital payment architecture enabling real-time P2P and P2B micro-transactions nationwide.
EMIs & Digital Banks	NayaPay, Sadapay, Digital Licenses	Electronic Money Institutions (EMIs) scaling multi-million user bases, coupled with newly issued full digital banking licenses to transition traditional banking to branchless, cloud-based models.